
Why Cross-Skeleton Retargeting Is Non-Identifiable: Structural Limits of Generative Motion Models

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Abstract

Cross-skeleton motion generation trains generative models to carry action structure and motion intention from one body to another. Yet an action-consistent target motion admits two observationally compatible explanations: source-preserving transfer and target-action recovery. We show that this ambiguity is structural rather than incidental: under standard generative objectives, the source-conditioned retargeting map is non-identifiable in sparse heterogeneous motion domains. Unpaired marginal matching yields *gauge non-identifiability*: a relative gauge between skeleton-specific latent spaces allows different source-conditioned maps to induce the same training evidence. Sparse paired supervision admits the complementary failure mode, *conditional-mean degeneration*: ambiguous action-level pairings drive squared-error objectives toward a target-action prototype that is independent of the source clip’s motion intention. To make the missing evidence observable, we introduce *Source-Instance Fidelity* (SIF), a diagnostic that tests whether the source-side geometry remains visible in the generated target motions after the target skeleton and action are fixed. Under this diagnostic, standard action-level success across animal motion domains recurrently coincides with behavior at the *source-blind floor*, while positive cases localize source-instance information to auxiliary motion-space constraints that narrow the observational equivalence class. The implication is that retargeting requires objectives and evaluations capable of identifying the source-conditioned map that a retargeting claim asserts. Project page: <https://keen-semolina-09811d.netlify.app/>.

1 Introduction

Motion generation is increasingly expected to work across bodies rather than within a single fixed skeleton. This expectation has grown from retargeting models that encode skeletal structure to generative and correspondence-based models designed for heterogeneous bodies [Aberman et al., 2020, Lee et al., 2023, Wang et al., 2024, Gat et al., 2025, Chen et al., 2025b]. A motion clip may come from a human, a quadruped, or an articulated animal, and a target body may have different proportions, joint topology, and kinematic limits. Cross-skeleton retargeting is the problem of generating a motion for the target body that keeps the action and the motion intention of the source clip, not merely a visually plausible motion on the target skeleton.

Most empirical evidence has been read through cross-skeleton contrastive retrieval under the standard protocol introduced with large heterogeneous motion libraries [Gat et al., 2025, Truebones Motions, 2025], measured by the area under the receiver-operating-characteristic curve (AUC), which summarizes a discrimination task rather than a retargeting claim. The generated target motion is treated as a query: target-library clips with the requested action are positives, clips with other actions are negatives, and the metric estimates the probability that a positive clip ranks above a randomly selected

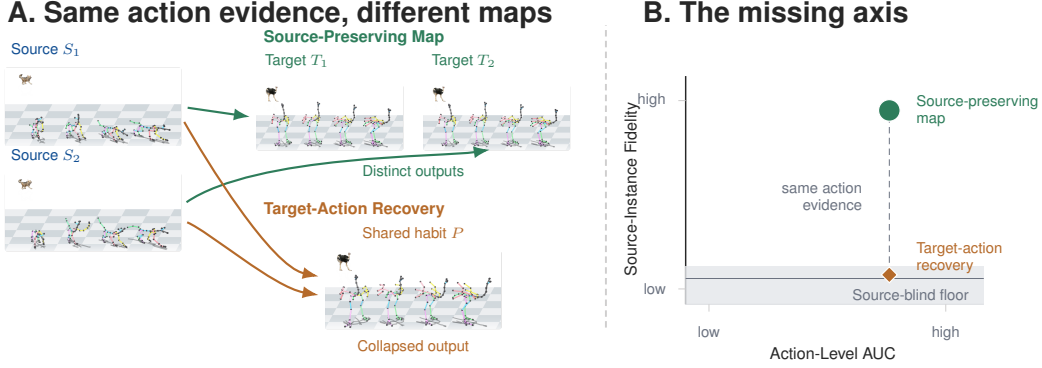


Figure 1: Action-level success does not identify the source-conditioned map. Two source clips with the same action can lead either to distinct target motions under a source-preserving map or to a shared target-action habit under source-blind recovery. Standard action-level AUC accepts both explanations; Source-Instance Fidelity tests whether the source-instance relation survives the change of body.

negative. This makes AUC an appropriate test of action recoverability in the target domain. It is silent, however, about whether the particular source clip selected the target motion that was produced.

The unsupported inference is the source-conditioned map. Retargeting claims that a particular source clip is mapped to a particular target motion, preserving timing and motion intention through the change of body. Yet an action-consistent output has two observationally compatible explanations: a source-preserving map may have transported the source instance, or a target-action habit may have produced a plausible motion for the requested body and action. Figure 1 summarizes this ambiguity. The action-level score accepts both explanations, while the missing source-instance question asks whether changing the source clip would change the target output accordingly.

This ambiguity becomes structural in sparse heterogeneous motion libraries. In the animal motion data we study, the Truebones zoo [Truebones Motions, 2025], motions span bodies with different skeletal topologies and joint counts: the overwhelming majority of body-and-action combinations are unobserved, and many of the observed ones contain only a single clip. Such data provide action labels and per-skeleton target examples, but not dense source-target correspondences that specify how a given source instance should move on a new body. The map asserted by retargeting is therefore not selected by the evidence that standard generative objectives consume.

The resulting limit is an identifiability limit, in the sense that the observed training evidence can be compatible with multiple latent explanations [Locatello et al., 2019, Khemakhem et al., 2020]. Under standard objectives in sparse heterogeneous domains, the source-conditioned retargeting map is *non-identifiable*. Per-skeleton marginal matching admits a relative gauge between skeleton-specific latent spaces, so distinct source-to-target maps can produce the same training evidence, a *gauge non-identifiability*. In the paired case, cell-paired squared-error regression under random target-cell pairing selects the target-action prototype rather than the source-conditioned motion, a *conditional-mean degeneration*. These mechanisms explain why better action recovery does not necessarily identify the retargeting map.

The missing observable is source-instance fidelity rather than target-action recognizability. We introduce *Source-Instance Fidelity* (SIF), a diagnostic that tests whether the source-side geometry remains visible in the generated target motions after the target skeleton and action are fixed. In controlled synthetic data, where the true correspondence is known, SIF separates source-preserving transport from source-blind generation. In sparse animal motion data, it reveals that action-level success can coexist with behavior near the *source-blind floor*.

Reading cross-skeleton generation through this lens reorganizes what counts as evidence. Action recovery and source-conditioned transfer are no longer collapsed into a single quantity, the two structural mechanisms above expose why standard objectives cannot select between them, and SIF supplies the source-instance signal that the action-level view never asks for. The same reading isolates a characterized positive case: ACE [Li et al., 2023], an adversarial cross-embodiment model that adds a motion-space constraint beyond per-skeleton distribution matching, ties its outputs back

74 to source-side structure tightly enough to narrow the observational equivalence class left open by
75 standard objectives.

76 The implication is that cross-skeleton retargeting cannot be evaluated only by asking whether the
77 target action is recognizable. A generated motion may be plausible, action-consistent, and retrievable
78 while still failing to express the motion intention of the source clip. What must be identified is the
79 source-conditioned map itself, and the central scientific question is which evidence and inductive
80 bias are sufficient to select it.

81 2 Related Work

82 **Cross-skeleton motion retargeting.** The retargeting literature has progressively weakened the
83 assumption that source and target bodies share a convenient representation. Early neural retargeting
84 made skeletal structure explicit, while later skeleton-agnostic representations and context-conditioned
85 sequence models sought motion spaces that remain meaningful across body graphs [Aberman et al.,
86 2020, Lee et al., 2023, Wang et al., 2024]. More recent work carries the same ambition into sparse
87 heterogeneous libraries, where generative models and sparse-correspondence procedures must operate
88 without dense paired examples [Gat et al., 2025, Kim et al., 2025, Li et al., 2023, Chen et al., 2025b].
89 Our concern is orthogonal to this modeling trajectory: once an output has the requested action, the
90 remaining question is whether the evidence identifies the particular source-conditioned map that
91 retargeting claims.

92 **Motion generation beyond fixed bodies.** Recent motion generation work also shows that the fixed
93 human skeleton is no longer the only natural domain of synthesis. The shared pressure across this
94 broader literature is to represent motion when body identity, morphology, and topology become part
95 of the conditioning problem rather than fixed background structure [Raab et al., 2024, Lee et al.,
96 2025, Egan et al., 2024, Chen et al., 2025a]. These settings make body variation a central modeling
97 object. The structural question here is complementary: when data are sparse and correspondences are
98 not instance-level, action labels and target examples may specify what action should appear on the
99 target body without specifying which source motion intention selected it.

100 **Identifiability and distributional transport.** Our formal argument is closest in spirit to identifi-
101 ability limits in representation learning, where marginal fit alone does not recover latent structure
102 without suitable inductive bias or auxiliary variables [Locatello et al., 2019, Khemakhem et al., 2020].
103 Distributional transport gives a second comparison point, since Schrödinger bridge objectives can
104 learn a coupling between marginals without observing instance-level correspondences [Bortoli et al.,
105 2021, Kim et al., 2024]. In cross-skeleton retargeting, however, the missing object is not merely a
106 transport between two marginal motion distributions. It is the relative alignment that tells a target
107 decoder how to interpret the source instance.

108 **Motion libraries and retrieval comparators.** Motion-library methods preserve realized target-side
109 motion snippets by construction, and this property connects classical motion graphs to modern sparse-
110 correspondence transfer [Kovar et al., 2002, Chen et al., 2025b]. This makes retrieval useful as a
111 diagnostic foil: it can expose what action-level evidence accepts without claiming to synthesize a new
112 source-conditioned motion. We use this distinction later to define a target-library comparator whose
113 role is diagnostic rather than generative. Appendix A gives additional context for these connections.

114 3 The Source-Conditioned Map

115 The object at stake is not a target motion distribution, but a map conditioned on a particular source
116 instance. Let $\mathcal{X}_s$ denote the motion space of skeleton s , with clips annotated by an action label c ;
117 this notation records that clips on different bodies occupy different kinematic domains. A retargeting
118 claim asserts a family of maps $T_{a \rightarrow b} : \mathcal{X}_a \rightarrow \mathcal{X}_b$ such that a source clip x_a on skeleton a is carried to
119 skeleton b while preserving the action, timing, and motion intention of that clip. This is a stronger
120 object than a conditional generator for plausible motions on b given the label c : the map must change
121 when the source instance changes, even when the action and target skeleton are held fixed.

122 Sparse heterogeneous motion libraries provide much weaker evidence than this map requires. In the
123 Truebones zoo used in our evaluation [Truebones Motions, 2025, Gat et al., 2025], the dataset contains

70 skeletons and 616 clips across 90 exact actions, yielding 6,300 possible skeleton-action cells. Only 413 of these cells are occupied, a fill rate of 6.6%, and the median occupied cell contains one clip. Thus a learner often sees that a skeleton can perform an action, but not multiple within-action instances from which the geometry of motion intention could be inferred, much less a dense set of source-target correspondences specifying how the same instance should appear on a different body.

This sparsity matters because source-instance evidence is relational. To ask whether a map preserves the source instance, one must be able to compare several source clips that share a source skeleton and action, generate the corresponding motions on the same target skeleton, and then ask whether the source-side geometry remains visible after retargeting. Truebones contains only 49 source-target-action triples with at least three source clips each; later paired comparisons are necessarily restricted further to the 37 triples that all evaluated methods can support. The structural question is therefore not whether action labels are useful, since they plainly are, but whether this sparse evidence selects a source-conditioned map rather than a target-action habit.

The distinction fixes the burden for the rest of the paper. Action-label evidence constrains which target motions are admissible for a requested action, whereas source-instance evidence constrains how differences among same-action source clips should reappear after the body changes. The former can support recognizable cross-skeleton generation; only the latter can identify the map that a retargeting claim asserts. Section 4 makes this gap precise by showing how standard objectives can leave multiple source-conditioned maps consistent with the same observed evidence.

4 Structural Non-Identifiability

A model can make plausible target motions while the objective remains indifferent to the source-conditioned map that produced them. This section makes the previous ambiguity structural by isolating two objective-level mechanisms: per-skeleton marginal matching leaves a relative latent coordinate freedom between bodies, while cell-paired squared-error regression under random target-cell pairing selects a target-action prototype. Both mechanisms preserve the evidence consumed by the objective while changing, or erasing, the source-instance dependence that retargeting requires.

The first mechanism appears whenever skeleton-specific encoders and decoders are coupled only through per-skeleton latent marginals. Let P_s be the motion distribution for skeleton s , let $\mu_s = (E_s)_\# P_s$ be the latent marginal, and write a shared-latent objective schematically as

$$L(E, D) = \sum_s L_s(D_s \circ E_s; P_s) + \Omega(\mu_1, \dots, \mu_K), \quad (1)$$

where Ω depends only on the per-skeleton latent marginals. For measurable invertible latent transformations $g = (g_1, \dots, g_K)$ with measurable inverses, define

$$G_\Omega(\mu) = \{g : \Omega((g_1)_\# \mu_1, \dots, (g_K)_\# \mu_K) = \Omega(\mu_1, \dots, \mu_K)\}. \quad (2)$$

For $g \in G_\Omega(\mu)$, set $E'_s = g_s \circ E_s$ and $D'_s = D_s \circ g_s^{-1}$. The relative transformations $g_b^{-1} \circ g_a$ are invisible to the marginal objective, but they are exactly the transformations that determine how a source latent is interpreted by a target decoder.

Theorem 1 (Gauge non-identifiability). *Let L be the objective in Eq. (1) with Ω depending only on per-skeleton latent marginals, and let $g \in G_\Omega(\mu)$ as in Eq. (2). Then $L(E', D') = L(E, D)$. If, in addition, there exist a, b such that $D_b \circ g_b^{-1} \circ g_a \neq D_b$ on a set of positive μ_a -measure, the induced retargeting maps differ: $T'_{a \rightarrow b} = D_b \circ g_b^{-1} \circ g_a \circ E_a \neq T_{a \rightarrow b} = D_b \circ E_a$ on a set of positive P_a -measure. $T_{a \rightarrow b}$ is determined by L only up to the relative gauge $g_b^{-1} \circ g_a$.*

This theorem is narrower than a generic impossibility result for disentanglement [Locatello et al., 2019] or nonlinear independent component analysis. It does not say that no auxiliary information can identify a representation. It says that the objective class above, when supplied with sparse skeleton-and-action evidence, does not select a relative latent alignment between skeletons. Identifiability under iVAE-style auxiliary variables [Khemakhem et al., 2020] requires the auxiliary distribution to induce full-rank variation in the prior sufficient statistics; on Truebones, where only 413 skeleton-action cells out of 6,300 are occupied and the median occupied cell contains one clip, skeleton-and-action metadata do not provide the empirical support needed to invoke that condition.

For Gaussian-aligned regularizers, the invariance contains $O(d)^K$; with $K = 70$ skeletons and the $d = 32$ latent dimension used for the finite-dimensional lower-bound calculation, quotienting the diagonal $O(d)$ subgroup, one common orthogonal factor applied to all skeletons, still leaves a lower bound of 34,224 continuous gauge degrees of freedom. This is not a count of distinct realized maps, because an observable retargeting ambiguity requires *decoder non-degeneracy*: the decoder must discriminate at least some directions on the gauge orbit. That condition is architecture-dependent. The checks in Appendix D record a small Kullback-Leibler (KL)-regularized VAE whose decoder nearly absorbs the orbit, with a gauge-to-noise ratio of 1.11, and a separate AnyTop hidden-state sweep showing that a transformer diffusion decoder for arbitrary skeleton topology [Gat et al., 2025] can make an orthogonal perturbation orbit visible after decoding. The AnyTop sweep is not the source of the $d = 32$ count; it is a decoder-observability boundary check. Orbit observability is nevertheless distinct from gauge-element selection: a decoder can distinguish rotation directions without the objective selecting which rotation aligns source and target latents. The main point is therefore not that every latent coordinate change is observable, but that the objective does not identify the relative latent alignment needed for a source-preserving map.

The second mechanism arises when sparse paired evidence is treated as if it were instance-level supervision. Consider a predictor trained with squared error from a source clip x_a and target skeleton b to a target motion x_b . If the target is drawn uniformly from the target skeleton and source-action cell rather than paired to the same motion instance, the risk is

$$L_{\text{paired}}(f) = \mathbb{E} [\|f(x_a, b) - x_b\|_2^2]. \quad (3)$$

The Bayes-optimal squared-error predictor is then the conditional mean of the target cell [Hastie et al., 2009]; in this regime, the conditional mean is a target-action prototype rather than a source-conditioned motion.

Proposition 1.1 (Conditional-mean degeneration). *Assume x_b is square-integrable, f ranges over measurable functions, and the empirical pairing draws targets uniformly at random from the (skeleton, action) cell $B(b, c) = \{x : (\text{skel}(x), \text{action}(x)) = (b, c)\}$ with $c = \text{action}(x_a)$. Then the (almost surely unique) minimiser of L_{paired} is $f^*(x_a, b) = |B(b, c)|^{-1} \sum_{x'_b \in B(b, c)} x'_b$, the empirical cell mean: a single fixed prototype per (target skeleton, source action), independent of the specific source clip.*

The proposition is deliberately scoped. It applies to cell-paired squared-error regression under random target-cell pairing, not to true instance-level pairs, which would supply exactly the correspondence the sparse regime lacks. It also assumes the target cell is nonempty and that motions are represented in the Euclidean vector space on which the squared-error loss is evaluated. The controlled ladder in Figure 2 shows the same mechanism in a setting where the correct source-conditioned transport T^* is known; the oracle variance is the within-cell variance of $T^*(x_a)$ over held-out source clips. As paired support per cell ranges from $M = 1$ to $M = 50$, the trained squared-error regressor preserves only 13% to 25% of that oracle within-cell output variance. No evaluated Truebones method exactly instantiates this random cell-pairing model. The closest real-data echo is AL-Flow, a label-conditional flow comparator that receives cluster and exact-action labels but no source motion. The latent-level diagnostic reported in Section 5 places AL-Flow at -0.001 , indicating that source-instance information can be lost before decoding when the training signal approximates cell-level pairing rather than instance-level correspondence. Appendix C contains the formal proofs: the gauge theorem in Appendix C.1, the conditional-mean proposition in Appendix C.2, and the retrieval corollary in Appendix C.3.

These two mechanisms also explain why retrieval comparators can be useful diagnostics without identifying the retargeting map. The following formal check records a boundary condition: a rule whose ranking depends only on features computed from joint positions, rather than from a learned shared latent, is outside the gauge variables of the objective above. It can therefore characterize the source-blind floor without selecting the source-conditioned map.

Corollary 1.1 (Retrieval is gauge-invariant). *Let $r : X_a \times \{b\} \rightarrow L_b$ be a retrieval rule that maps a source to a clip in the target-skeleton library by a function of a feature ϕ that is itself gauge-invariant, for example kinematic descriptors computed directly from joint positions rather than from a learned latent. Then r is invariant under the gauge group G_Ω of any shared-latent objective: replacing E_a with $g_a \circ E_a$ leaves $r(x_a, b)$ unchanged. Furthermore $r(x_a, b)$ is a single existing clip rather than an average, so the conditional-mean operator of Prop. 1.1 does not apply.*

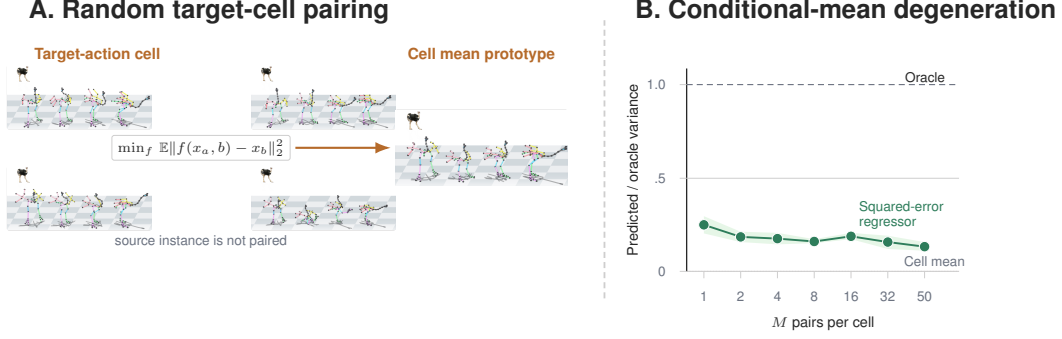


Figure 2: Cell-paired squared-error regression collapses toward the target-cell prototype. Panel A illustrates random target-cell pairing: several admissible target clips define a cell, but the squared-error optimum is the cell mean prototype rather than a source-conditioned motion. Panel B reports the controlled ladder with known source-conditioned transport T^* ; from $M = 1$ to $M = 50$ pairs per cell, predicted output variance remains only 13% to 25% of oracle within-cell variance $\text{Var}(T^*(x_a))$. The shaded band is the three-seed standard deviation.

225 The corollary is a measurement statement, not a constructive recommendation, and its invariance
 226 follows from the fact that the score is computed in motion space rather than through the learned latent
 227 coordinates. We later instantiate this measurement role through label-only baselines and ANCHOR,
 228 our deterministic target-library retrieval comparator. These comparators can mark what action-level
 229 evidence accepts at the source-blind floor precisely because they avoid the two objective mechanisms
 230 analyzed here. They do not identify the retargeting map; they expose why standard evidence alone
 231 cannot.

232 5 Source-Instance Fidelity and Empirical Evidence

233 The missing observable is whether source variation remains ordered after retargeting. A source-
 234 conditioned map should not merely produce a plausible target motion; after the source skeleton,
 235 target skeleton, and action are fixed, it should preserve the relative geometry among source clips.
 236 We therefore define *Source-Instance Fidelity* (SIF) on each eligible triple (a, b, c) . For source clips
 237 $x_i, x_j \in \mathcal{X}_a$ and generated target motions $y_i, y_j \in \mathcal{X}_b$, each pair is cropped to its common duration
 238 and put into a common comparison frame by least-squares Procrustes superposition, which removes
 239 global translation, rotation, and isotropic scale [Goodall, 1991]. Let $d_{\text{src}}(i, j)$ and $d_{\text{out}}(i, j)$ be the
 240 aligned per-frame discrepancies on the source and output sides. Using Pearson’s product-moment
 241 correlation [Pearson, 1896], we define

$$\rho_{\text{SIF}} = \text{corr}(\{d_{\text{src}}(i, j)\}_{i < j}, \{d_{\text{out}}(i, j)\}_{i < j}), \quad (4)$$

242 within each source-target-action triple and report the macro-average over triples, with confidence
 243 intervals obtained by bootstrap resampling.

244 SIF asks a different question from action retrieval. A generator can recover the right target action
 245 by returning a target-action habit, in which case the action-level score may be positive while ρ_{SIF}
 246 remains near zero. A source-preserving map should instead make larger source-side changes appear
 247 as larger target-side changes after target skeleton and action are fixed. For failure-mode analysis, we
 248 also compute *latent Source-Instance Fidelity* (L-SIF) before decoding, replacing output distances
 249 with Frobenius distances between the corresponding target latents [Golub and Van Loan, 1996]. This
 250 locates whether source-instance information is absent in the latent representation or erased by the
 251 decoder.

252 The synthetic calibration fixes the scale of the diagnostic. In the controlled 2×2 setting, the source-
 253 conditioned transport T^* is known for every source clip, providing the instance-level correspondence
 254 absent from the real corpus. Appendix F shows that the oracle sits near one, while two source-blind
 255 references are centered near zero with upper bootstrap bounds below $+0.14$. We therefore use
 256 $|\rho_{\text{SIF}}| < 0.10$ as a conservative shorthand for behavior inside the source-blind floor.

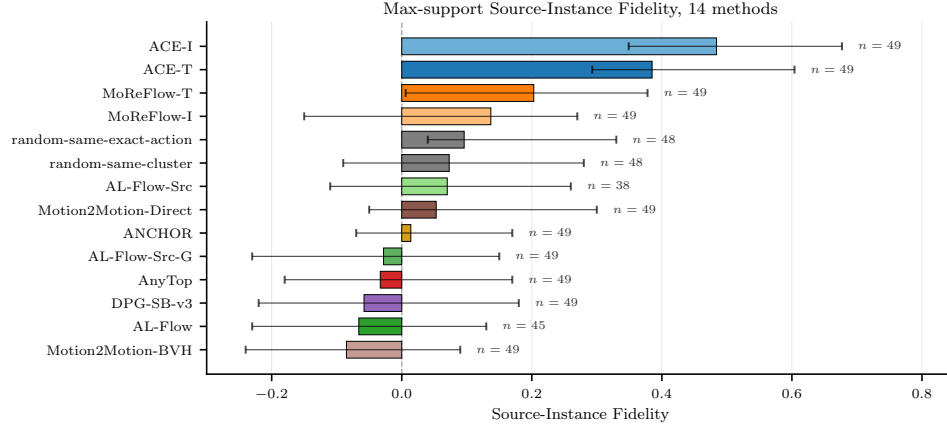


Figure 3: Source-Instance Fidelity on each evaluated row’s largest available Truebones support. The point is the macro-average SIF over eligible source-target-action triples, the interval is a bootstrap confidence interval over triples, and the printed n is the number of triples supporting that estimate. On max support ($n = 38$ to 49), most evaluated rows remain at the source-blind floor. ACE-I and ACE-T are the nontrivial positive rows highlighted by the SIF axis; label-only rows are interpreted by the action-level audit below.

On the real corpus, SIF must be reported with the support that sparse data allow. The largest Truebones SIF evaluation set contains 49 eligible triples, and the common support across all rows contains 37 triples. We use each row’s largest available support for its primary estimate and reserve the 37-triple intersection for paired comparisons. The comparison is organized by evidentiary role rather than by rank: published generative objectives are evaluated beside rows that isolate sparse correspondence, supervision-matched labels, latent transport, target-library retrieval, and label-only sampling [Gat et al., 2025, Kim et al., 2025, Li et al., 2023, Chen et al., 2025b, Bortoli et al., 2021, Kim et al., 2024]. Appendix B records source-conditioning adaptations and split conventions.

ANCHOR occupies a distinct diagnostic role: it is not a retargeting generator. Given a source clip, target skeleton, and action request, it predicts a coarse source action cluster from motion-space descriptors and retrieves an existing target-library clip using that prediction, exact action metadata, and direct kinematic descriptors. Because this rule sits outside the learned latent objectives analyzed in Section 4, it can show when action-level evidence and source-conditioned transfer separate. The scoring rule appears in Appendix G.

The main empirical pattern is that action-consistent generation usually does not preserve source-instance geometry. Figure 3 shows the largest support each evaluated row admits: seven learned rows, ANCHOR, and the two random label-only references all fall inside $|\rho_{\text{SIF}}| < 0.10$ on their max-support estimates, giving ten of fourteen evaluated rows. On the 37-triple common support, the support shift raises random-same-exact-action to a label-only positive ($\rho = +0.186$) rather than a learned source-preserving exception, while ACE-I is robust under clustered bootstrap and ACE-T separates from the floor under iid, source-skeleton, and target-skeleton bootstrap but is sensitive to action clustering. MoReFlow-T remains borderline, with lower confidence bounds of -0.003 on the 37-triple support and $+0.018$ on max support, so we do not treat it as a stable exception. The result is the empirical counterpart of the structural claim in Section 4: action-level evidence can be present while the source-conditioned map remains unidentified. Appendix E reports the corresponding support, bootstrap, and latent-SIF checks.

The qualitative panorama makes the same ambiguity visible. In Figure 4, a single Bird $\rightarrow$ KingCobra attack query is held fixed while the evaluated methods produce visibly different target motions. The figure is not a substitute for SIF, but it shows why a visual action judgment alone cannot identify the source-conditioned map: many outputs can look action-consistent while corresponding to different maps.

This view also clarifies what the quantitative diagnostic is, and is not, claiming. The panorama does not ask the reader to choose a visually best method, nor does it treat a single attractive output as

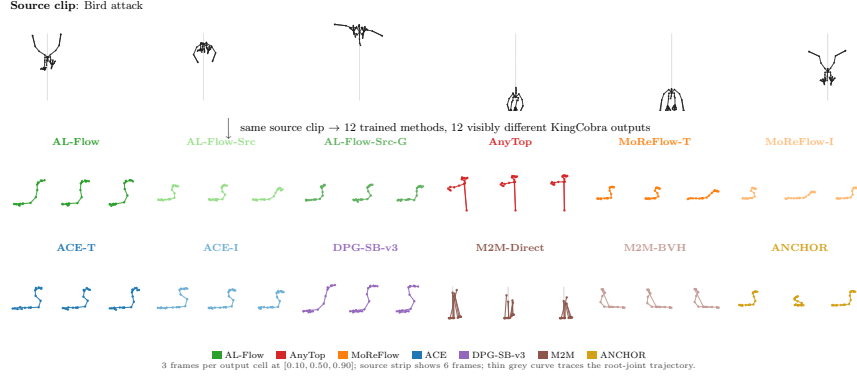


Figure 4: Qualitative non-identifiability panorama. A single Bird $\rightarrow$ KingCobra attack query is shown across evaluated methods. All rows receive the same source clip, target skeleton, and action request, yet the generated target motions differ visibly across methods. The figure is qualitative evidence for the ambiguity isolated by SIF: action-consistent target motion can be produced without identifying a unique source-conditioned map.

evidence of transfer. It fixes the query and exposes the equivalence class that remains after action recognizability has been satisfied. SIF then supplies the missing relational test: if the source clip changes within the same source-target-action cell, the target output should change in the corresponding way rather than merely remain a plausible KingCobra attack.

The standard score says yes whenever the target action is recoverable, even when SIF says the source-conditioned map has not been selected. Table 1 gives the numerical audit, and Figure 5 visualizes the same comparison. The random-same-cluster reference chooses a random target clip with the same coarse action cluster, while random-same-exact-action chooses a random target clip with the exact action label whenever such a clip exists. These rules have no access to a source-conditioned retargeting map, yet random-same-cluster matches ANCHOR within ± 0.005 on both the overall and held-out splits. The exact-action variant exceeds ANCHOR by $+0.119$ overall and $+0.250$ held-out, but only on the 221/600 overall and 57/200 held-out coverage where exact-action positives exist in the target library. The qualifier is part of the claim: strong action-level evidence can be generated by label availability alone, without identifying the map. Appendix H gives the broader enumeration and action-level comparison tables behind this audit.

Table 1: Label-only retrieval can match or exceed ANCHOR’s action-level AUC without solving the retargeting problem. AUC is measured under the same cluster-tier Procrustes protocol used for ANCHOR. The exact-action reference has restricted coverage, so its larger gaps are only defined on the rows where the target library contains an exact-action candidate.

Method	Overall $n/600$	Held-out $n/200$	Overall AUC	Held-out AUC
ANCHOR	600	200	0.764	0.673
random-same-cluster	570	177	0.769 (+0.005)	0.678 (+0.005)
random-same-exact-action	221	57	0.883 (+0.119)	0.923 (+0.250)

Parentetical Δ values are method AUC minus ANCHOR. random-same-cluster matches ANCHOR within ± 0.005 on both splits. random-same-exact-action exceeds ANCHOR by $+0.119$ overall and $+0.250$ held-out, but only on the 221/600 and 57/200 coverage where exact-action positives exist; the coverage qualifier travels with the gap.

The label-only audit changes how the positive SIF rows should be read. Once action-level AUC can be reproduced by references with no source-conditioned map, the question is not whether a method also scores well on action recovery, but whether it contains independent motion-space evidence that preserves the source instance after the target skeleton and action are fixed. On the 37-triple cross-method intersection, where every evaluated method is scored on the same triples, ACE-T and ACE-I are the retargeting rows that remain above the source-blind floor; paired tests further distinguish ACE-I from both the random-same-exact-action reference and MoReFlow-T, the strongest nontrivial

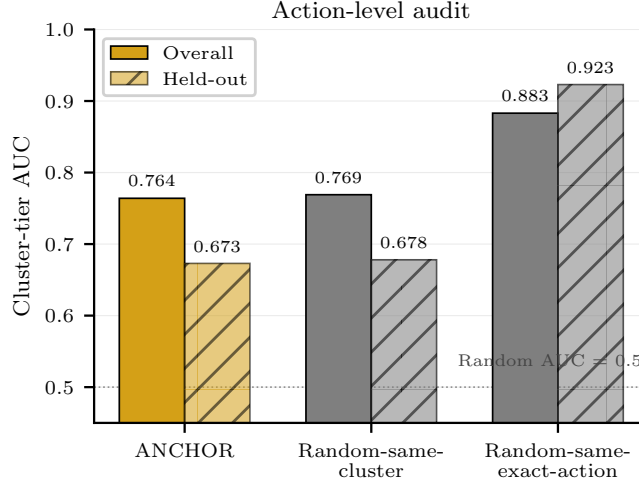


Figure 5: Action-level AUC cannot distinguish source-conditioned transport from label-based retrieval. ANCHOR retrieves from the target library using action information and motion descriptors, while the two random references use only label constraints. Their AUC values remain in the same action-level regime, showing that the standard score can accept target-action recovery even when source-instance transfer has not been established. The exact-action reference is coverage-limited to 221/600 overall and 57/200 held-out queries.

alternative. Thus ACE is not a counterexample to the action-level critique. It is the characterized case in which an auxiliary motion-space constraint supplies evidence outside the standard score.

The ACE controls in Appendix I localize that evidence to the motion-space adversarial loss rather than to source copying. Table 19 and Figure 19 show that removing this loss and retraining the variant across three seeds lowers SIF to a mean of 0.214, with range [0.175, 0.276], below the ACE baselines and close to the source-blind regime. The nearest-neighbor leakage check in Table 20 gives the complementary control: 76 to 79 percent of ACE outputs are closer, under coarse kinematic features, to a target-skeleton training clip than to the source clip, which does not support a direct source-copying explanation. The structural lesson is therefore specific: auxiliary motion-space objectives can help select a source-preserving element from the equivalence class, while action-level AUC by itself cannot certify that selection.

6 Conclusion

Cross-skeleton retargeting can appear successful under action-level evaluation while the source-conditioned map remains unidentified. The analysis isolates two structural mechanisms behind this ambiguity. *Gauge non-identifiability* leaves per-skeleton latent spaces related by an unresolved gauge under per-skeleton marginal matching, while *conditional-mean degeneration* makes cell-paired squared-error regression under random target-cell pairing recover a target-action prototype rather than the motion intention of the source clip. Source-Instance Fidelity then makes the missing evidence measurable by asking whether source-side geometry remains visible after target skeleton and action are fixed.

The empirical picture follows the structural analysis. In sparse heterogeneous animal motion data, action-consistent outputs often remain at the source-blind floor, while label-only references show why action-level AUC can stay positive without selecting a map. ACE is the characterized positive case in our study, because its auxiliary motion-space constraint preserves enough source-instance evidence to narrow the observational equivalence class. The broader implication is constructive: progress in retargeting should be judged by whether an objective improves action recovery, source-conditioned transport, or both. The open scientific question is which auxiliary motion-space objectives select a source-preserving map from the observational equivalence class identified here.

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421 A Extended Related Work

422 Cross-skeleton retargeting sits between character animation, heterogeneous motion generation, and
 423 representation learning. Earlier retargeting work established that motion can be shared across bodies
 424 by making skeletal structure explicit, by learning motion spaces intended to be less dependent on
 425 topology, or by conditioning sequence models on skeleton context [Aberman et al., 2020, Lee et al.,
 426 2023, Wang et al., 2024]. The generative setting studied here inherits this ambition, but places it in
 427 sparse heterogeneous libraries where dense source-target correspondences are absent [Gat et al., 2025,
 428 Kim et al., 2025, Li et al., 2023, Chen et al., 2025b]. Our distinction is not that these methods fail
 429 to synthesize plausible target motion. It is that standard evidence can accept a target motion whose
 430 action is correct while the source-conditioned map remains unidentified.

431 Recent work on broader object and animal motion further clarifies why this identification question
 432 matters. Together, these papers move motion synthesis away from one canonical human body and
 433 toward settings in which body identity, morphology, and topology become part of the generative
 434 problem itself [Raab et al., 2024, Lee et al., 2025, Egan et al., 2024, Chen et al., 2025a]. These works
 435 broaden the range of bodies that motion models must handle. The structural question in this paper is
 436 complementary: in sparse heterogeneous data, which evidence selects the motion intention carried
 437 from the source clip rather than a target-side action habit?

Table 2: Truebones zoo regime. The SIF diagnostic is evaluated only on triples with at least three source clips, while the cross-method intersection keeps the triples scored by all evaluated methods.

Quantity	Value	Evidence
Skeletons	70	§1.1
Total clips	616	§1.1
Coarse clusters	10	§1.1
Exact actions	90	§1.1
Skeleton-action cells	6,300 (70×90)	§1.1
Occupied cells	413 (6.6%)	§1.1
Median clips per occupied cell	1	§1.1
SIF eligibility (≥ 3 source clips per triple)	49 triples (171 queries)	§5.1.0
SIF cross-method intersection	37 triples (130 queries)	§5.1.0

The formal argument is closest in spirit to identifiability limits for unsupervised representation learning. Locatello et al. [Locatello et al., 2019] show that unsupervised disentanglement cannot be guaranteed without inductive bias, while the iVAE analysis of Khemakhem et al. [Khemakhem et al., 2020] shows how suitable auxiliary variables can make nonlinear independent components identifiable. Our gauge theorem is narrower than either statement. It does not rule out identification with stronger side information; it shows that per-skeleton marginal matching does not select the relative latent alignment required by a source-preserving retargeting map.

Retrieval and motion-library methods provide a useful foil because they operate directly in motion space. Motion graphs [Kovar et al., 2002] already exploit the idea that a library can preserve physically realized motion snippets, while sparse-correspondence retrieval shows how such libraries can be used when source and target bodies do not share topology [Chen et al., 2025b]. These methods motivate the diagnostic comparison used in the main text: retrieval can characterize what action-level evidence accepts, but by itself it does not identify the source-conditioned retargeting map.

Schrödinger bridge and flow objectives offer another route through the same space of ambiguities. Diffusion Schrödinger bridges [Bortoli et al., 2021] and unpaired neural Schrödinger bridges [Kim et al., 2024] motivate transport between distributions, and flow matching is one way modern retargeting methods instantiate this distributional view [Kim et al., 2025]. In the sparse cross-skeleton setting, distributional transport alone does not remove the relative-gauge ambiguity. This is why the paper separates action-level transport evidence from the source-instance evidence measured by SIF.

B Data Regime and Method Roster

The structural limits studied in the main paper arise in a sparse heterogeneous motion library. Table 2 records the Truebones regime used throughout the experiments. The important fact is not merely that the library is small, but that most skeleton-action cells are empty and most occupied cells contain a single clip, making instance-level correspondence rare even when action labels are available.

The evaluated roster in Table 3 separates published generative models, supervision-matched comparators, the deterministic ANCHOR comparator, and source-blind random references. The public method names in this table are the only names used for result interpretation.

Source-conditioning adaptations. Two rows require additional care because their original public form does not match the SIF input contract. AnyTop is unconditional with respect to a source clip, so the evaluated row adds a self-supervised source-motion encoder with four learned motion queries and injects the resulting latent through cross-attention into the target-skeleton decoder; training remains unpaired self-reconstruction on Truebones. Motion2Motion is a sparse-correspondence retrieval-and-blending procedure rather than a learned generator, so we report both Motion2Motion-Direct, a reimplementation on the 13-channel representation, and Motion2Motion-BVH, the official BVH implementation, using auto-heuristic correspondences and excluding evaluator candidate clips from the target example pool. These details affect only how each row receives source information; SIF is computed by the same source-target-action protocol across rows.

Table 3: Method roster. Train-skeleton counts distinguish transductive evaluation on all skeletons from inductive evaluation with held-out target skeletons.

Method	Family	Train skeletons
AnyTop	source-conditioned diffusion	70
ACE-T	motion-space adversarial	70
ACE-I	motion-space adversarial	60
MoReFlow-T	flow matching	70
MoReFlow-I	flow matching	60
AL-Flow	label-conditional flow, no source motion	60
AL-Flow-Src	label-conditional flow + source motion + skeleton id	60
AL-Flow-Src-G	label-conditional flow + source motion + skeleton graph	60
Motion2Motion-Direct	paired bone regression	60
Motion2Motion-BVH	paired BVH regression	60
DPG-SB-v3	latent Schrödinger bridge	60
ANCHOR	gauge-invariant retrieval (comparator)	60
random-same-cluster	trivial baseline	—
random-same-exact-action	trivial baseline	—

The AnyTop row uses the source-conditioned extension required by SIF; the original generator is unconditional with respect to a source clip. AL-Flow denotes the label-conditional flow family. ANCHOR is a deterministic gauge-invariant retrieval comparator, not a retargeting solution. DPG-SB-v3 is evaluated in its latent-bridge form without decoded-motion penalties (see Appendix K).

C Proofs and Formal Details

C.1 Proof of Theorem 1

Proof. For each skeleton s and motion $x \in \mathcal{X}_s$, the reparameterized encoder-decoder pair satisfies

$$D'_s(E'_s(x)) = D_s(g_s^{-1}(g_s(E_s(x)))) = D_s(E_s(x)),$$

because g_s is a measurable bijection. The per-skeleton reconstruction term $L_s(D_s \circ E_s; P_s)$ is therefore unchanged pointwise.

The latent marginal induced by the reparameterized encoder is $(E'_s)_\# P_s = (g_s \circ E_s)_\# P_s = (g_s)_\# \mu_s$. Since $g \in G_\Omega(\mu)$, the marginal regularizer also remains unchanged:

$$\Omega((g_1)_\# \mu_1, \dots, (g_K)_\# \mu_K) = \Omega(\mu_1, \dots, \mu_K).$$

Combining the unchanged reconstruction terms with the unchanged regularizer gives $L(E', D') = L(E, D)$.

For the retargeting map, define

$$A = \{z : D_b(g_b^{-1}(g_a(z))) \neq D_b(z)\}.$$

By the decoder non-degeneracy hypothesis in Theorem 1, this set has positive μ_a -measure for some pair a, b . Let $B = E_a^{-1}(A) \subseteq \mathcal{X}_a$. Since $(E_a)_\# P_a = \mu_a$, $P_a(B) = \mu_a(A) > 0$. For every $x \in B$,

$$T'_{a \rightarrow b}(x) = D_b(g_b^{-1}(g_a(E_a(x)))) \neq D_b(E_a(x)) = T_{a \rightarrow b}(x),$$

so the two retargeting maps differ on a set of positive P_a -measure. The objective therefore determines the map only up to the relative gauge $g_b^{-1} \circ g_a$. $\square$

Corollary 1.2 (Orthogonal-gauge lower bound under exact Gaussian alignment). *Suppose $\Omega(\mu_1, \dots, \mu_K) = \sum_s \text{KL}(\mu_s \| \mathcal{N}(0, I_d))$, or any rotation-invariant function of the marginals, and each $\mu_s = \mathcal{N}(0, I_d)$. Then G_Ω contains $O(d)^K$, the product of orthogonal groups. The full product has $K d(d-1)/2$ continuous parameters. After quotienting the diagonal $O(d)$ subgroup, one common orthogonal factor applied to all skeletons, the relative gauge retains $(K-1)d(d-1)/2$ continuous degrees of freedom.*

495 *Proof.* If $R \in O(d)$ and $z \sim \mathcal{N}(0, I_d)$, then $Rz \sim \mathcal{N}(0, RR^\top) = \mathcal{N}(0, I_d)$. Thus every per-
 496 skeleton choice $(R_1, \dots, R_K) \in O(d)^K$ preserves all Gaussian-aligned marginals and belongs to
 497 G_Ω whenever Ω is rotation-invariant. Since $O(d)$ has dimension $d(d-1)/2$, the product group
 498 has $K d(d-1)/2$ continuous parameters. Quotienting the diagonal subgroup removes one $O(d)$
 499 factor, because a common orthogonal transformation applied to all skeletons cancels in the relative
 500 transformation $g_b^{-1} \circ g_a$, leaving $(K-1)d(d-1)/2$ relative-gauge parameters. For the setting
 501 used in the paper, $K = 70$ and $d = 32$, giving $69 \cdot 32 \cdot 31/2 = 34,224$ relative-gauge degrees of
 502 freedom in this orthogonal subgroup. This is a lower bound on relative-gauge degrees of freedom,
 503 not by itself a count of distinct decoded retargeting maps; that latter count also depends on decoder
 504 non-degeneracy. $\square$

505 C.2 Proof of Proposition 1.1

506 *Proof.* Condition on a source clip x_a and target skeleton b , and write $c = \text{action}(x_a)$. Under
 507 the proposition’s sampling rule, the target x_b is drawn uniformly from the finite cell $B(b, c) =$
 508 $\{x : (\text{skel}(x), \text{action}(x)) = (b, c)\}$, independently of the source instance once the action cell is
 509 fixed. For any proposed prediction $u = f(x_a, b)$, the sampling rule gives $x_b \perp x_a \mid (b, c)$, hence
 510 $\mathbb{E}[x_b \mid x_a, b] = \mathbb{E}[x_b \mid b, c]$. Therefore

$$\mathbb{E}\|u - x_b\|_2^2 = \|u - \mathbb{E}[x_b \mid b, c]\|_2^2 + \mathbb{E}\|x_b - \mathbb{E}[x_b \mid b, c]\|_2^2.$$

511 The second term is independent of u , so the finite-cell minimizer is unique; in the population version
 512 the same equality gives the almost surely unique conditional-mean predictor $u = \mathbb{E}[x_b \mid b, c]$. In the
 513 empirical finite-cell case, this conditional expectation is

$$|B(b, c)|^{-1} \sum_{x'_b \in B(b, c)} x'_b.$$

514 It depends on the target skeleton and source action, but not on the specific source clip. The Bayes
 515 predictor is therefore the target-action cell mean. $\square$

516 C.3 Proof of Corollary 1.1

517 *Proof.* The retrieval rule r depends on the source motion through a feature $\phi(x_a)$ that is computed
 518 directly from motion space and is invariant to the latent-gauge transformations in G_Ω . Replacing E_a
 519 by $g_a \circ E_a$ therefore changes no argument supplied to r , and the returned library clip is unchanged.
 520 Since $r(x_a, b)$ is a single existing target-skeleton clip rather than an average over target-cell motions,
 521 the conditional-mean operator in Proposition 1.1 is not the relevant limiting operation. $\square$

522 D Decoder Non-Degeneracy Boundary

523 The gauge theorem requires a decoder that can make at least part of the gauge orbit observable. In
 524 a small Kullback-Leibler (KL)-regularized VAE, random gauge rotations and matched-magnitude
 525 noise remain near the noise floor in decoded effects, with a gauge-to-noise ratio of 1.11. This is the
 526 boundary case in which the decoder nearly absorbs the orbit.

527 A separate AnyTop transformer diffusion hidden-state sweep tests decoder observability, not the
 528 $d = 32$ gauge-count model. Across hook layers, reverse timesteps, source motions, and target
 529 skeletons, all 72 tested cells, formed by four skeletons, two source clips, three hook layers, and three
 530 reverse timesteps, give rotation-induced output divergence above the matched-noise control, with
 531 a minimum gauge-to-noise ratio of 1.71. This check does not instantiate the 34,224-dimensional
 532 lower-bound calculation. It shows only that decoder observability can hold in a real transformer
 533 diffusion motion model, leaving gauge-element selection as the structural problem. Figure 6 and
 534 Table 4 summarize these rotation-versus-noise perturbation checks.

535 The cross-seed latent checks in Tables 5 and 6 give a complementary view. Orthogonal Procrustes
 536 alignment [Goodall, 1991] explains a substantial fraction of cross-seed latent variation, while effective
 537 rank separates rich latent variation from low-dimensional collapse. These checks defend the boundary
 538 of the theory: decoder observability can hold even when the objective does not select the relative
 539 gauge. Figures 7, 8, and 9 visualize the same distinction between latent alignment, representation
 540 richness, and source-instance fidelity.

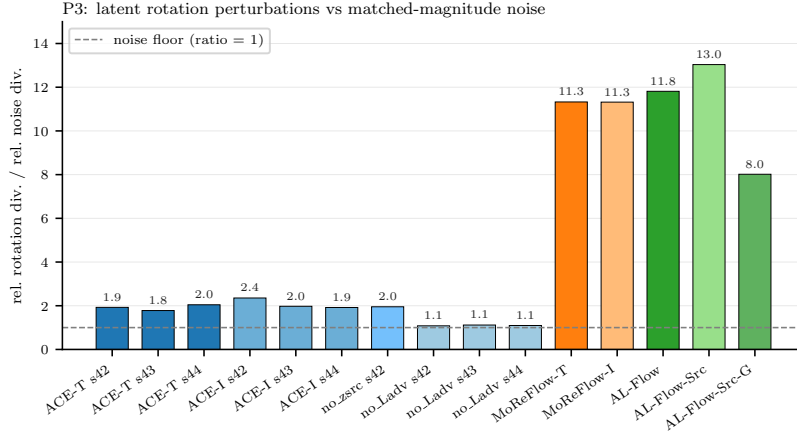


Figure 6: Decoder non-degeneracy check. Rotation-induced changes are compared with matched-magnitude noise in the latent space; ratios above one indicate that the decoder distinguishes at least some directions on the gauge orbit.

Table 4: Rotation versus noise perturbations on the SIF-intersection queries. The ACE ablation with the motion-space adversarial loss removed is the boundary case near the noise floor, while the other evaluated methods make the orbit visible after decoding.

Method	Seed	rel. div. rotation	rel. div. noise	Ratio	Verdict
ACE-T (3 seeds)	42	0.240	0.213	1.93	non-degenerate
	43	0.219	0.183	1.78	non-degenerate
	44	0.209	0.171	2.04	non-degenerate
ACE-I (3 seeds)	42	0.121	0.089	2.35	non-degenerate
	43	0.128	0.071	1.98	non-degenerate
	44	0.141	0.087	1.92	non-degenerate
No z_{src} ablation	42	0.117	0.083	1.95	non-degenerate
No L_{adv} ablation (3 seeds)	42	1.113	1.154	1.08	at noise floor
	43	0.810	0.849	1.12	at noise floor
	44	0.699	0.734	1.10	at noise floor
MoReFlow-T	–	0.113	0.017	11.33	non-degenerate (strong)
MoReFlow-I	–	0.107	0.016	11.32	non-degenerate (strong)
AL-Flow	–	0.052	0.007	11.81	non-degenerate (strong)
AL-Flow-Src	–	0.065	0.009	13.04	non-degenerate (strong)
AL-Flow-Src-G	–	0.048	0.008	8.02	non-degenerate (strong)

Each row reports the mean over 130 SIF-intersection queries with 20 random $O(d)$ rotations + 20 matched-magnitude noise samples per query. The no- L_{adv} ablation is statistically indistinguishable from the noise floor (Wilcoxon $p > 0.4$ for all three seeds); every other method exceeds the noise floor at $p < 10^{-12}$.

541 E Source-Instance Fidelity Robustness

542 SIF is a diagnostic for whether source-side geometry remains visible after the target skeleton and
543 action are fixed. The main text reports the qualitative pattern; Tables 7 and 8 give the max-support
544 and intersection estimates behind that pattern.

545 The robustness checks ask whether the same ordering survives dependence across source skeletons,
546 target skeletons, and actions. Table 9 reports clustered bootstrap intervals, and Table 10 reports paired
547 comparisons on the same 37 triples. Figure 10 visualizes the clustered intervals. The conclusion is
548 the same as in the main text: ACE-I and ACE-T are the robust nontrivial positives on the intersection,
549 random-same-exact-action is a label-only positive, and MoReFlow-T remains borderline.

Table 5: Orthogonal Procrustes alignment across seeds. The table reports how much latent variation can be explained by an $O(d)$ alignment across the SIF-intersection queries.

Group	Seed pair	Frac. $O(d)$ -explained	Row-cosine
ACE-T	s42 vs s43	0.522	0.893
ACE-T	s42 vs s44	0.703	0.900
ACE-T	s43 vs s44	0.585	0.905
ACE-I	s42 vs s43	0.689	0.935
ACE-I	s42 vs s44	0.712	0.928
ACE-I	s43 vs s44	0.775	0.941
no_L_adv	s42 vs s43	0.482	0.866
no_L_adv	s42 vs s44	0.428	0.847
no_L_adv	s43 vs s44	0.753	0.881

Frac. $O(d)$ -explained is the fraction of cross-seed latent variance accounted for by the optimal $O(d)$ Procrustes rotation. *Row-cosine* is the mean per-row cosine similarity after alignment. Both are computed on per-target-skeleton latents across 130 SIF-intersection queries, aggregated over 21 target skeletons.

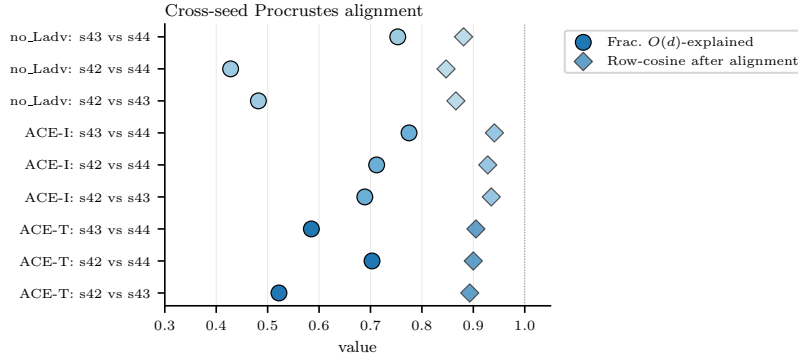


Figure 7: Cross-seed Procrustes alignment. High explained fractions indicate that latent spaces differ partly by approximately orthogonal transformations, matching the relative-gauge mechanism in Theorem 1.

550 The eligibility rule also matters for interpretation. Table 11 and Figure 11 show that SIF triples have
551 at least three source clips by construction, but only a few triples provide more than three. Figure 12
552 checks that the main pattern is not a monotone artifact of the diversity ratio R .

553 Latent SIF (L-SIF) repeats the SIF computation before decoding, when a method exposes a latent
554 representation on the relevant queries. Table 12 and Figure 13 show that latent preservation and
555 decoded SIF can decouple. This motivates the failure-mode taxonomy used in the qualitative
556 discussion: a method may collapse in latent space, collapse after decoding, or recover output variation
557 without proving source memorization.

558 F Synthetic Calibration

559 The synthetic 2×2 environment calibrates SIF in a setting where the source-conditioned transport
560 is known. Figure 14 visualizes the three reference regimes used in Section 5: the oracle transport
561 lies near one, while two source-blind references lie near zero. Table 13 records the corresponding
562 numerical calibration.

563 The same environment also isolates conditional-mean degeneration. The cell-paired squared-error
564 predictor receives random target samples from the correct target skeleton and action cell, but not
565 instance-level pairs. As Table 14 shows, increasing the number of random cell-pairs does not recover
566 the oracle within-cell variance.

Table 6: Effective-rank diagnostic on latent distributions. Higher effective rank indicates richer latent variation, but rank alone does not imply that the source-conditioned map has been identified.

Method	Sample shape	Spectral flatness	Effective rank / 256
no_L_adv s42	$3,376 \times 256$	0.0000	1.29
ACE-T s42	$3,376 \times 256$	0.0000	2.13
ACE-I s42	$3,376 \times 256$	0.0019	3.00
AL-Flow	$1,040 \times 256$	0.0101	5.84
MoReFlow-I	$3,823 \times 256$	0.0593	12.84
MoReFlow-T	$3,823 \times 256$	0.0631	14.99
AnyTop encoder z	$13,000 \times 256$	0.1049	41.41

Effective rank computed on the per-method latent distribution over the 130 SIF-intersection queries.
AnyTop’s encoder hidden state is an order of magnitude richer than the Stage-A bottleneck latents.

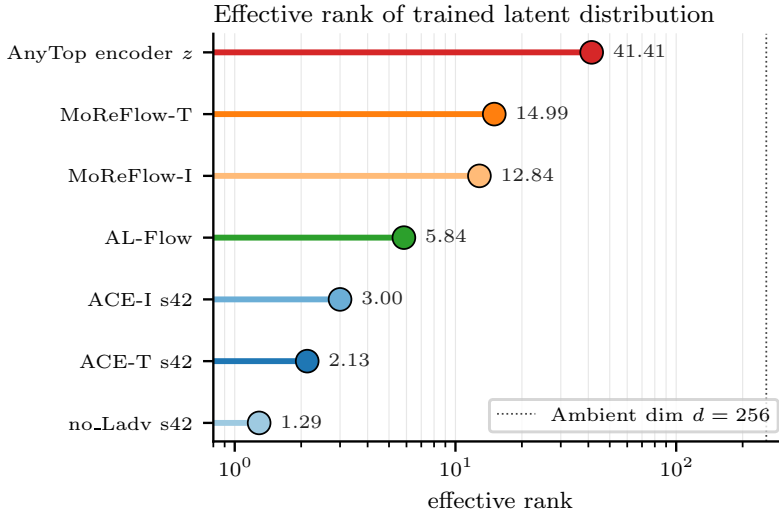


Figure 8: Effective rank across latent representations. The diagnostic separates representation richness from source-instance fidelity.

567 G Deterministic Comparator and Motion Descriptors

568 ANCHOR is included to make the diagnostic boundary explicit. It is not a generator and does not
569 synthesize a new target motion. For a query consisting of a source clip, a target skeleton, and an
570 action request, ANCHOR first predicts a coarse source action cluster using a random-forest classifier
571 [Breiman, 2001] trained on motion-space descriptors from the training skeletons. It then searches the
572 target skeleton’s motion library and returns one existing clip.

573 The motion descriptor, denoted $Q(x)$, is computed from joint-position trajectories rather than from a
574 learned latent representation. It summarizes center-of-mass displacement and speed, heading velocity,
575 contact timing, cadence, dominant limb usage, and global clip statistics. The candidate pool is the
576 union of all target-skeleton clips in the predicted cluster and the ten clips with largest Q -descriptor
577 similarity to the source. Candidate clips are scored by cluster agreement, descriptor similarity,
578 and exact-action agreement, using the fixed comparator weights $(w_{\text{cluster}}, w_Q, w_{\text{action}}) = (1, 2, 3)$
579 recorded in the method roster.

580 This construction is why ANCHOR is gauge-invariant in the sense of Corollary 1.1: changing a
581 learned latent parametrization cannot change a score computed directly from center-of-mass, heading,
582 contact, cadence, and limb-usage measurements on the joint trajectories. At the same time, because
583 ANCHOR returns an existing library clip, it does not claim to identify the source-conditioned
584 retargeting map. Its purpose is to mark a diagnostic boundary, showing that action recovery can

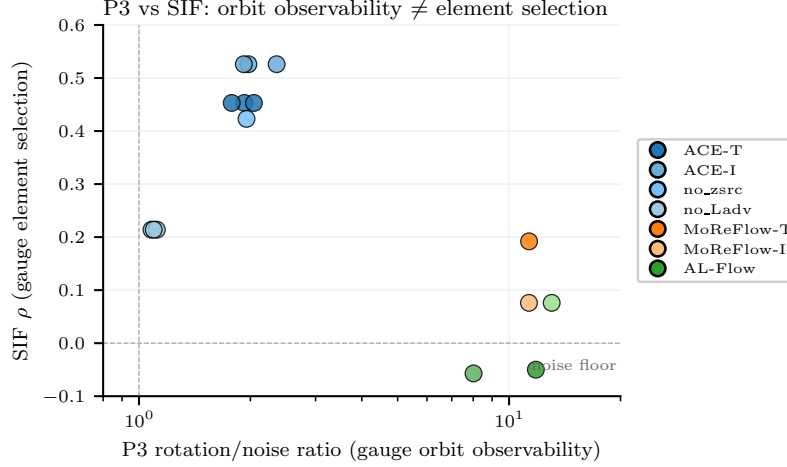


Figure 9: Decoder non-degeneracy and SIF are distinct. A method may expose latent perturbations after decoding while still lying near the source-blind floor under SIF.

Table 7: Max-support SIF results. The table reports each method on the largest set of eligible triples it can score, together with the corresponding 37-triple intersection interval.

Method	ρ_{SIF} (max)	n	R	95% CI on intersection
ACE-I	0.484	49	0.11	0.526 [0.349, 0.677]
ACE-T	0.385	49	14.94	0.453 [0.293, 0.604]
MoReFlow-T	0.203	49	0.56	0.192 [0.006, 0.378]
MoReFlow-I	0.137	49	0.56	0.076 [-0.150, 0.270]
AL-Flow-Src	0.070	38	0.22	0.076 [-0.110, 0.260]
Motion2Motion-Direct	0.053	49	2.42	0.131 [-0.050, 0.300]
ANCHOR	0.014	49	0.17	0.040 [-0.070, 0.170]
AL-Flow-Src-G	-0.028	49	0.18	-0.057 [-0.230, 0.150]
AnyTop	-0.033	49	3.01	-0.010 [-0.180, 0.170]
DPG-SB-v3	-0.058	49	0.58	-0.019 [-0.220, 0.180]
AL-Flow	-0.066	45	0.16	-0.050 [-0.230, 0.130]
Motion2Motion-BVH	-0.085	49	1.31	-0.067 [-0.240, 0.090]
Random baselines:				
random-same-exact-action	0.096	48	0.36	0.186 [0.040, 0.330]
random-same-cluster	0.073	48	0.83	0.104 [-0.090, 0.280]

Identity reference: $\rho_{\text{SIF}} = +1.000$. Cell-mean reference: $\rho_{\text{SIF}} = 0.000$. The ρ column reports each method’s maximal-support estimate (n in the third column); the 95% CI column is from the cross-method 37-triple intersection bootstrap. **Bold** marks methods whose intersection lower CI exceeds +0.01 (*robust positives*); MoReFlow-T at +0.006 is treated as borderline and is not bolded (see `tab_sif_intersect`).

585 remain high at the source-blind floor. Table 15 records the action-cluster predictor, and Figure 15
586 shows how ANCHOR matches vary across action clusters.

587 H Action-Level AUC and Label Recovery

588 The standard cross-skeleton contrastive retrieval AUC is an action-level ranking test: it asks whether
589 a generated target motion ranks positives above negatives in a target library. Section 5 shows that
590 this test can accept label-only behavior on the covered queries. The appendix gives the broader
591 enumeration and comparison tables behind that statement. Table 16 and Figure 16 report the full
592 enumeration, while Figure 17, Table 17, Table 18, and Figure 18 collect the restricted-query and
593 AUC-band controls.

Table 8: SIF on the 37-triple cross-method intersection. This is the common support used for clustered intervals and paired comparisons.

Method	ρ_{SIF} (95% CI)	R (95% CI)	Status
ACE-I	0.526 [0.349, 0.677]	0.10 [0.04, 0.18]	robust
ACE-T	0.453 [0.293, 0.604]	4.59 [0.36, 11.06]	robust
MoReFlow-T	0.192 [0.006, 0.378]	0.53 [0.34, 0.77]	borderline
random-same-exact-action	0.186 [0.040, 0.330]	0.32 [0.15, 0.57]	robust
Motion2Motion-Direct	0.131 [-0.050, 0.300]	2.05 [1.50, 2.71]	
random-same-cluster	0.104 [-0.090, 0.280]	0.74 [0.45, 1.11]	
MoReFlow-I	0.076 [-0.150, 0.270]	0.57 [0.33, 0.90]	
AL-Flow-Src	0.076 [-0.110, 0.260]	0.23 [0.14, 0.34]	
ANCHOR	0.040 [-0.070, 0.170]	0.20 [0.06, 0.40]	
AnyTop	-0.010 [-0.180, 0.170]	2.98 [1.78, 4.62]	
DPG-SB-v3	-0.019 [-0.220, 0.180]	0.58 [0.33, 0.92]	
AL-Flow	-0.050 [-0.230, 0.130]	0.10 [0.06, 0.15]	
AL-Flow-Src-G	-0.057 [-0.230, 0.150]	0.19 [0.11, 0.30]	
Motion2Motion-BVH	-0.067 [-0.240, 0.090]	1.32 [0.91, 1.81]	

$n = 37$ cross-method intersection. Identity reference: $\rho = +1.000$. Cell-mean reference: $\rho = 0.000$. **Status** marks the lower-CI position relative to zero: *robust* if lower CI $> +0.01$; *borderline* if lower CI is in $(0, +0.01]$ (MoReFlow-T at $+0.006$).

Table 9: Clustered bootstrap intervals for SIF. Clustering by source skeleton, target skeleton, or action tests whether the positive conclusions are driven by a narrow subset of triples.

Method	iid	cluster: source-skel	cluster: target-skel	cluster: action
ACE-I	0.526 [0.351, 0.681]	0.526 [0.323, 0.709]	0.526 [0.365, 0.671]	0.526 [0.242, 0.629]
ACE-T	0.453 [0.293, 0.602]	0.453 [0.284, 0.618]	0.453 [0.274, 0.605]	0.453 [-0.009, 0.619]
MoReFlow-T	0.192 [-0.003, 0.380]	0.192 [-0.023, 0.386]	0.192 [-0.031, 0.405]	0.192 [-0.039, 0.357]
random-same-exact-action	0.185 [0.046, 0.326]	0.185 [0.069, 0.304]	0.185 [0.027, 0.354]	0.185 [0.000, 0.272]
Motion2Motion-Direct	0.131 [-0.061, 0.317]	0.131 [-0.098, 0.385]	0.131 [-0.075, 0.307]	0.131 [-0.720, 0.265]
random-same-cluster	0.104 [-0.095, 0.298]	0.104 [-0.057, 0.279]	0.104 [-0.115, 0.315]	0.104 [-0.272, 0.189]
AL-Flow-Src	0.076 [-0.108, 0.264]	0.076 [-0.109, 0.269]	0.076 [-0.112, 0.248]	0.076 [-0.222, 0.169]
MoReFlow-I	0.076 [-0.132, 0.279]	0.076 [-0.132, 0.271]	0.076 [-0.167, 0.319]	0.076 [-0.108, 0.474]
ANCHOR	0.040 [-0.068, 0.157]	0.040 [-0.078, 0.176]	0.040 [-0.068, 0.167]	0.040 [0.000, 0.046]
AnyTop	-0.010 [-0.185, 0.166]	-0.010 [-0.191, 0.166]	-0.010 [-0.207, 0.185]	-0.010 [-0.253, 0.142]
DPG-SB-v3	-0.019 [-0.215, 0.178]	-0.019 [-0.208, 0.188]	-0.019 [-0.244, 0.228]	-0.019 [-0.487, 0.045]
AL-Flow	-0.050 [-0.231, 0.137]	-0.050 [-0.187, 0.106]	-0.050 [-0.259, 0.161]	-0.050 [-0.239, 0.183]
AL-Flow-Src-G	-0.057 [-0.239, 0.130]	-0.057 [-0.207, 0.087]	-0.057 [-0.261, 0.128]	-0.057 [-0.523, 0.022]
Motion2Motion-BVH	-0.067 [-0.237, 0.110]	-0.067 [-0.247, 0.116]	-0.067 [-0.221, 0.089]	-0.067 [-0.320, 0.142]

Bold lower CI excludes zero. Bootstrap $N = 10,000$, seed = 42. ACE-I is the only method whose lower CI excludes zero under all four bootstrap schemes; ACE-T and random-same-exact-action survive source-skel and target-skel clusterings but action-clustered drops their lower CI to or below zero due to action concentration in the 37-triple intersection.

594 I ACE Controls

595 ACE is the characterized positive case in this study because its motion-space constraint narrows the
596 observational equivalence class more effectively than the other evaluated objectives. The ablation
597 in Table 19 and Figure 19 supports this interpretation: removing the motion-space adversarial term
598 reduces SIF across three seeds, while the full ACE-T and ACE-I models remain the robust nontrivial
599 positives on the intersection.

600 The nearest-neighbor leakage check in Table 20 addresses a different explanation: that ACE could be
601 lifting SIF by memorizing source clips. Most ACE outputs are closer to a target-pool training clip
602 than to the corresponding source clip in the reported four-dimensional kinematic feature space, which
603 weakens source memorization as the explanation for the SIF lift.

604 J Qualitative Evidence and Supplementary Videos

605 The qualitative figures are not used as quantitative proof, but they make the SIF failure modes visible.
606 The appendix provides three full-grid still figures that correspond to the supplementary videos. Each

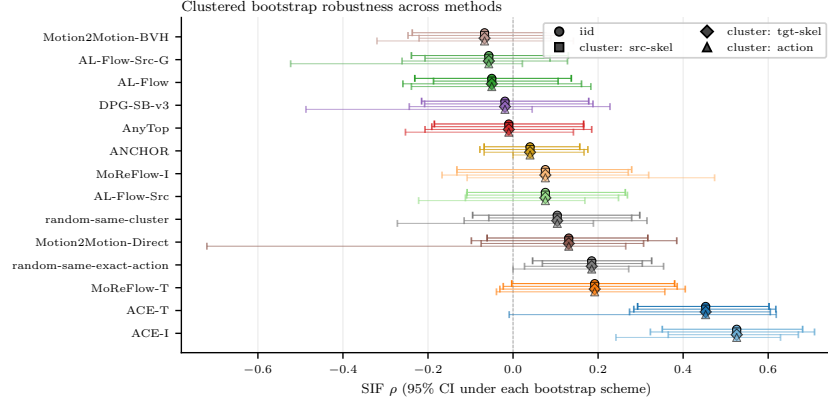


Figure 10: Clustered confidence intervals for SIF. The figure visualizes the dependence-aware intervals reported in Table 9.

Table 10: Paired SIF comparisons on the 37-triple intersection. Paired tests compare methods on identical triples rather than across different supports.

Pairing	Mean $\Delta\rho$	Wilcoxon p	Bootstrap 95% CI on $\Delta\rho$	Excludes zero
ACE-I vs random-same-exact-action	+0.340	4.35×10^{-3}	[+0.127, +0.541]	yes
ACE-T vs random-same-exact-action	+0.268	7.94×10^{-3}	[+0.047, +0.477]	yes
ACE-I vs MoReFlow-T (next non-trivial)	+0.334	7.21×10^{-3}	[+0.089, +0.582]	yes
ACE-T vs MoReFlow-T	+0.262	5.27×10^{-2}	[-0.004, +0.533]	no (borderline)
ACE-T vs no_L_adv (3-seed mean)	+0.153	6.10×10^{-3}	—	yes
ACE-I vs no_L_adv (3-seed mean)	+0.312	1.70×10^{-3}	—	yes

All paired tests on the 37-triple intersection ($n = 37$). ACE-I separates from both the strongest source-blind reference (random-same-exact-action) and the next non-trivial generative method (MoReFlow-T); ACE-T separates from random-same-exact-action but is borderline against MoReFlow-T.

grid fixes a source-target triple and compares the evaluated methods across three source clips, so rows that move in near lockstep indicate source-blind behavior while rows with source-dependent variation indicate partial source conditioning. Figure 4 gives the compact method panorama in the main text. Figures 20, 21, 22, and 23 provide the appendix still-image views corresponding to this qualitative protocol.

The supplementary material contains nine companion videos: two main-body videos for the panorama and failure-mode figures, and seven appendix videos for additional source-target triples. The videos use the same projection rule as the still figures and synchronize playback phase across methods, making source-blind lockstep behavior and source-conditioned variation easier to see.

K Latent Schrödinger Bridge Comparator Scope

The evaluated latent Schrödinger bridge comparator, reported under the public method name DPG-SB-v3 in the roster, uses retrieval-initialized noise, a shared conditioned bridge, a target-side latent constraint, and a latent-space cycle term. The decoded-motion bone, contact, and smoothness penalties were not included in this evaluation. This caveat is important for interpretation: DPG-SB-v3 tests where the evaluated latent bridge variant lies relative to the source-blind floor, not an exhaustive set of motion-space penalties for bridge objectives.

L Potential Societal Impacts

Cross-skeleton motion generation may broaden access to motion content in animation, education, assistive design, and scientific visualization by reducing the need to record every action separately for every body. In these settings, a source-preserving model could help creators transfer motion intention across characters, support low-resource motion libraries, and make comparative studies of movement

Table 11: Pair counts per SIF triple. The minimum and median values are fixed by the eligibility rule; the few larger cells determine how much within-cell geometry is available.

Method	Min pairs	Median pairs	Max pairs	n triples
ACE-I	3	3	5	49
ACE-T	3	3	5	49
AL-Flow	3	3	5	45
AL-Flow-Src	3	3	5	38
AL-Flow-Src-G	3	3	5	49
ANCHOR	3	3	5	49
AnyTop	3	3	5	49
DPG-SB-v3	3	3	5	49
Motion2Motion-BVH	3	3	5	49
Motion2Motion-Direct	3	3	5	49
MoReFlow-I	3	3	5	49
MoReFlow-T	3	3	5	49
random-same-exact-action	3	3	5	48
random-same-cluster	3	3	5	48

Min and median are uniformly 3 because the SIF benchmark requires ≥ 3 source clips per triple by design; max reflects the few triples in Truebones with 4 or 5 source clips. Variation across methods comes from the few queries that some baselines fail to score.

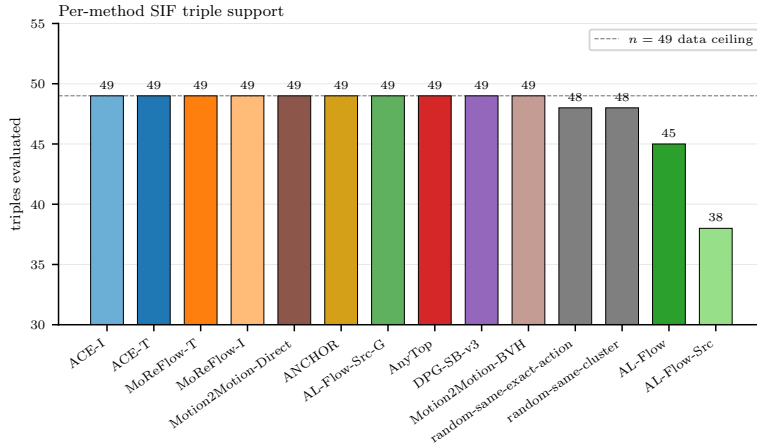


Figure 11: Pair-count distribution for SIF-eligible triples.

628 easier. This paper contributes a diagnostic and theoretical account of when such transfer claims are
629 not identified by standard evidence, rather than a deployed generator.

630 The same capability also carries risks. More reliable motion transfer could make synthetic embodied
631 media easier to produce, including misleading performances attributed to people or characters who
632 did not enact them, and action-level evaluation may overstate reliability when the source-conditioned
633 map is not actually preserved. Biases in the source and target motion libraries may also be propagated
634 across bodies, especially when some anatomies, actions, or movement styles are sparsely represented.
635 Our analysis supports a conservative evaluation stance: cross-skeleton methods should report source-
636 instance diagnostics alongside action-level scores, disclose the motion domains and licenses on which
637 they are trained, and avoid claims of retargeting fidelity when only target-action recovery has been
638 established.

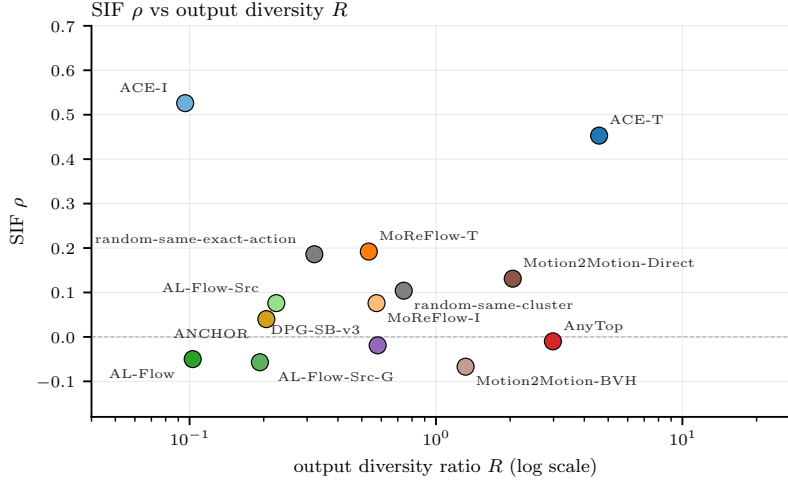


Figure 12: SIF versus output diversity ratio. High output diversity is not sufficient for source-instance preservation, because the geometry must align with the source-side geometry after the target skeleton and action are fixed.

Table 12: SIF and L-SIF failure-mode taxonomy. L-SIF is a latent-space diagnostic and is therefore reported only where the representation is comparable across queries.

Failure mode	Method	SIF ρ	L-SIF ρ
Source-conditional	MoReFlow-T	0.192	0.165
Output-only recovery	ACE-I (3 seeds avg)	0.503	0.083
	ACE-T (3 seeds avg)	0.408	0.088
	no_z_src	0.423	0.107
Mixed (both modest)	no_L_adv (3 seeds avg)	0.214	0.055
	AL-Flow-Src	0.076	0.090
Decoder collapse	MoReFlow-I	0.076	0.132
	AL-Flow-Src-G	-0.057	0.109
	AnyTop	-0.010	0.141
Latent collapse	AL-Flow	-0.050	-0.001

Cross-method Pearson correlation between SIF and L-SIF: $r = -0.003$, $p = 0.99$ across $n = 16$ method runs. Per-method L-SIF confidence intervals individually cross zero, so the four-mode classification is observational across method families rather than a per-method significance claim.

Table 13: Synthetic SIF calibration. The oracle lies near perfect source-instance preservation, while source-blind references remain near zero.

Configuration	ρ_{SIF} (95% CI)	R
Oracle (output = $T^*(x_a)$)	0.9957 [0.9714, 1.0000]	1.16
Random clip from cell	-0.0002 [-0.0870, 0.1382]	19.60
Random Gaussian noise	-0.0009 [-0.1225, 0.1286]	25.70

Synthetic 2×2 paired-dense setting, $n = 336$ pairs. The oracle saturates SIF; both source-blind references sit at zero.

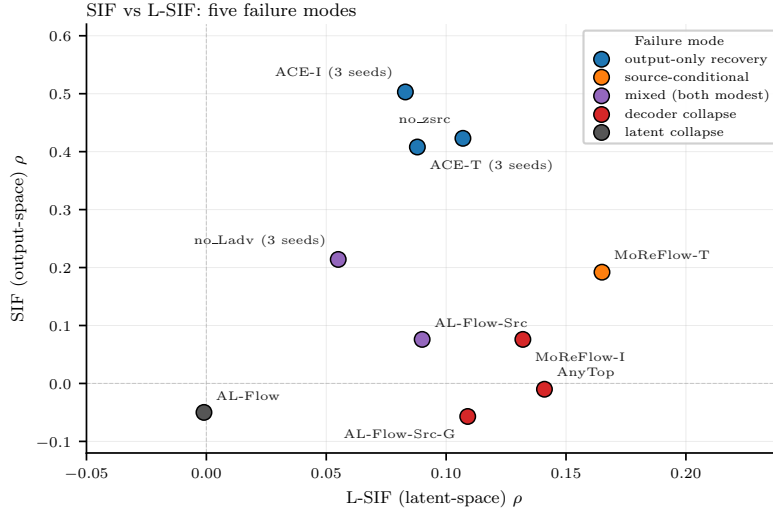


Figure 13: SIF and L-SIF decoupling. Source-instance fidelity in decoded motion need not coincide with a positive latent-space correlation.

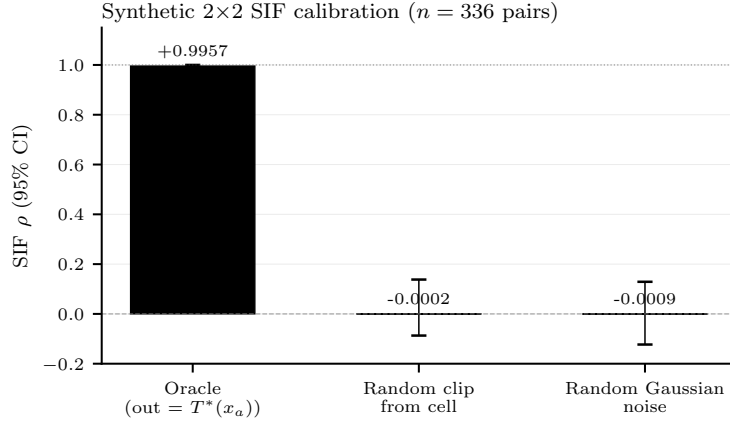


Figure 14: Synthetic 2×2 SIF calibration. The oracle uses the known source-conditioned transport $T^*(x_a)$; the random-clip and Gaussian references deliberately ignore the source instance.

Table 14: Conditional-mean degeneration in the controlled 2×2 setting. The predicted variance remains a small fraction of oracle variance even when the number of random target-cell pairs increases.

M pairs / cell	$\sigma_{\text{pred}}^2 / \sigma_{\text{oracle}}^2$	$\text{MSE} / \sigma_{\text{oracle}}^2$	n_{seeds}
1	0.249 ± 0.047	2.32 ± 2.40	3
2	0.185 ± 0.028	0.99 ± 0.10	3
4	0.175 ± 0.031	0.93 ± 0.02	3
8	0.162 ± 0.011	0.93 ± 0.03	3
16	0.188 ± 0.018	0.92 ± 0.03	3
32	0.158 ± 0.035	0.95 ± 0.09	3
50	0.133 ± 0.023	1.00 ± 0.06	3

Across all M and 3 seeds, the variance ratio $\sigma_{\text{pred}}^2 / \sigma_{\text{oracle}}^2$ stays at 0.13–0.25 even at $M = 50$ pairs per cell, confirming prop: cm predicted cell-mean compression.

Table 15: ANCHOR action-cluster predictor. The held-out row shows that the cluster prediction itself is imperfect, which is why ANCHOR is interpreted as a comparator rather than an oracle.

Domain	Cluster classifier accuracy	n correct / total
In-domain (train skeletons)	99.8%	—
Held-out (reserved cross-skeleton)	61.5%	59 / 96

Replacing the filename cluster with the predicted cluster pulls Procrustes performance below random on exact-tier and on the B1 hand-curated subset. The in-domain row reports only the percentage from the original training log; the corresponding numerator/denominator was not preserved and is therefore listed as *not available*.

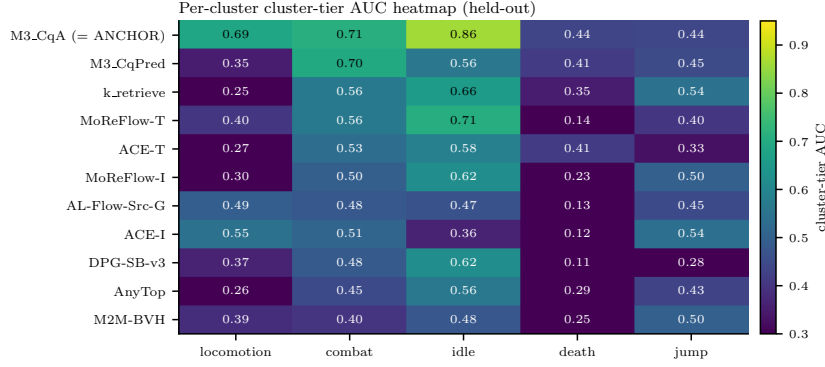


Figure 15: Per-cluster ANCHOR matches. The comparator’s behavior varies across action clusters, reinforcing that it characterizes a floor rather than a universal retargeting rule.

Table 16: Full ANCHOR enumeration across cluster-eligible triples. The held-out subsets show that strong action-level ranking can arise from target-library label structure without identifying a source-conditioned map.

Subset	n triples (cluster-eligible)	Cluster-tier AUC (95% CI)
All	30,029	0.824 [0.820, 0.829]
In-distribution (same train pair)	21,838	0.845 [0.841, 0.849]
Mixed (one held-out skeleton)	7,622	0.774 [0.762, 0.787]
Held-out (both cross-skeleton)	569	0.711 [0.669, 0.758]

ANCHOR on the 30,497-pair full enumeration. Cluster-eligibility filter removes triples whose target skeleton has no in-cluster positives. The mixed subset is the load-bearing held-out generalisation case.

Table 17: Action-level AUC band for generated variants and supervision-matched comparators. These values are not interpreted as source-preserving transfer; they show how much action-level evidence can be explained without SIF.

Variant	Conditioning channels	Cluster AUC	Band
AnyTop	self-supervised source-conditioned diffusion	0.466	inside
Precursor	action-conditioned generation	0.484	inside
Precursor	latent-bridge generator	0.484	inside
DPG-SB-v3	latent bridge objective; no decoded-motion penalties	0.448	inside
AL-Flow	cluster + exact-action labels	0.538	inside
AL-Flow-Src	AL-Flow + source motion + skel. id	0.542	inside
AL-Flow-Src-G	AL-Flow + source motion + skel. graph	0.470	inside

Support-qualified band is [0.39, 0.59] for the seven listed configurations. The two precursor rows are reported only for the action-level AUC-band analysis and are not part of the 14-method SIF roster. The wider [0.37, 0.55] strict-support band covers all 13 evaluated configurations (see `tab_master_auc` for the full 14-method comparison).

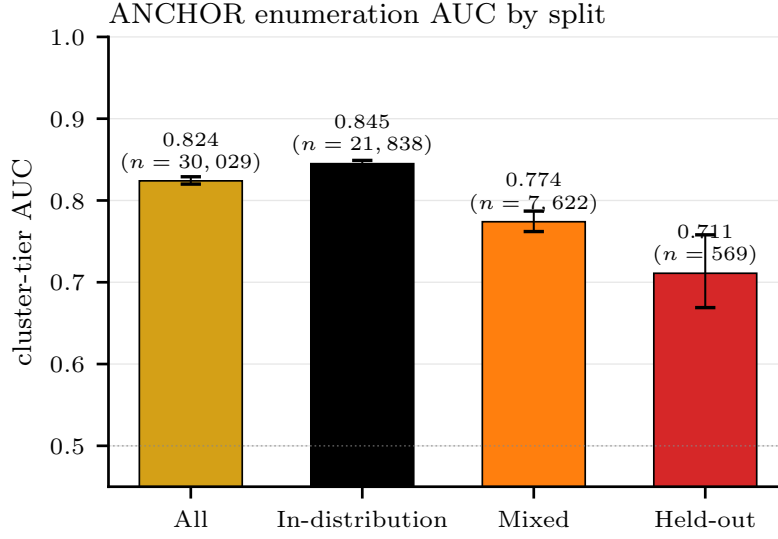


Figure 16: Enumeration breakdown for ANCHOR. The large enumeration separates in-distribution, mixed, and held-out skeleton regimes.

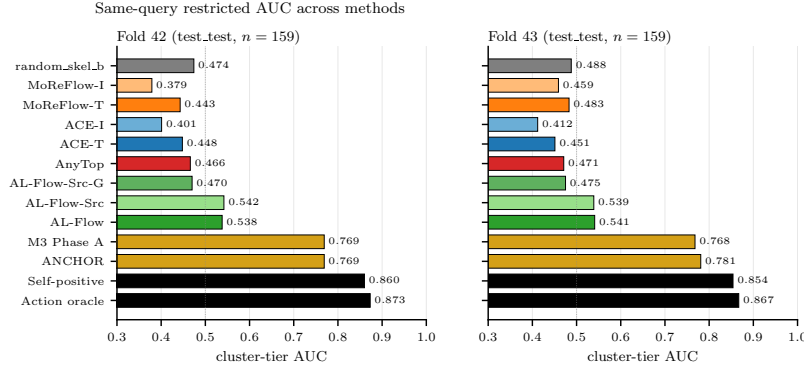


Figure 17: Same-query restricted AUC comparison. Restricting the comparison to shared queries keeps the contrast focused on the action-level ranking signal.

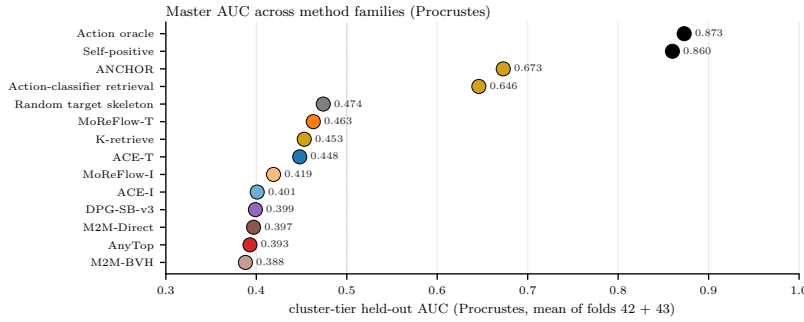


Figure 18: Action-level AUC band. Many methods occupy a narrow action-ranking band, which motivates SIF as the missing source-instance axis.

Table 18: Master action-level AUC comparison on held-out queries. The table reports cluster-tier AUC under three distances and separates library-based references from generative methods.

Method	Cluster-tier held-out AUC (average of folds 42 and 43)		
	Procrustes	Z-DTW	Q-comp
Action oracle	0.873	0.846	0.894
Self-positive reference	0.860	0.847	0.909
ANCHOR	0.673	0.707	0.718
Cluster-Classifer Retrieval	0.646	0.669	0.693
Q-Retrieval (Q-feature only)	0.453	0.523	0.597
MoReFlow-T	0.463	0.485	0.510
MoReFlow-I	0.419	0.426	0.524
ACE-T	0.448	0.451	0.529
ACE-I	0.401	0.469	0.550
Random target skeleton (null)	0.474	0.510	0.530

Motion2Motion-BVH has Procrustes only and is reported separately. MoReFlow-T and MoReFlow-I Procrustes values are the per-fold-corrected averages from EVIDENCE DOSSIER § 9; an earlier internal report had stale values 0.443 and 0.379.

Table 19: ACE ablation controls. Removing the motion-space adversarial loss lowers SIF, while the feature-loss ablation diverges and therefore yields no usable checkpoint.

Variant	Seed	ρ_{SIF}	95% CI	R	Status
ACE-T (full model)	42	0.453	[0.293, 0.604]	4.59	reference
ACE-I (full model)	42	0.526	[0.349, 0.677]	0.10	reference
No z_{src} input	42	0.423	[0.235, 0.581]	0.05	trained, 50k steps
No L_{feat}	—	—	—	—	training diverged (NaN at step 4,600)
No L_{adv}	42	0.175	[−0.028, 0.373]	1.43	trained, 50k steps
No L_{adv}	43	0.191	[0.001, 0.377]	0.59	trained, 50k steps
No L_{adv}	44	0.276	[0.069, 0.465]	0.49	trained, 50k steps
No L_{adv} (mean)	3 seeds	0.214	—	0.84	$\sigma = 0.054$

Three independent retrains for the no- L_{adv} ablation. Removing the motion-space adversarial loss collapses ACE’s source-instance fidelity across all three seeds (paired Wilcoxon $p < 0.01$ vs both ACE-T and ACE-I baselines). The L_{feat} ablation diverges during training and yields no usable checkpoint.

Table 20: ACE nearest-neighbor leakage check. Distances are computed in a kinematic feature space; the majority of ACE outputs are closer to the target pool than to the source clip.

Method	Queries	Closer-to-target-pool	Mean d_{src}	Mean d_{pool}	Ratio
ACE-I	130	103 (79.2%)	3.449	2.215	1.56
ACE-T	130	99 (76.2%)	3.697	2.650	1.39

Distance computed in a four-dimensional kinematic feature space (centre-of-mass displacement, centre-of-mass variance, mean velocity, foot-contact density). 76–79% of ACE outputs are closer to a target-pool training clip than to the corresponding source clip, weakening source-memorisation as an explanation of ACE’s SIF lift.

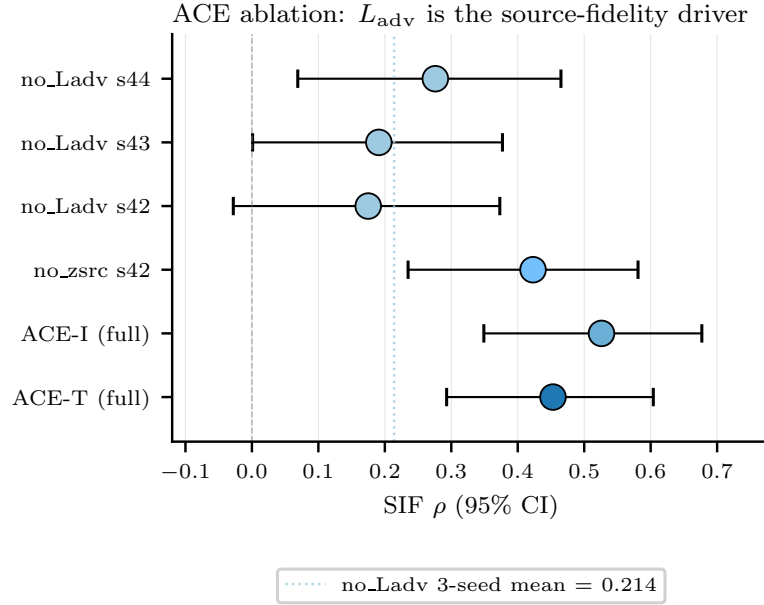


Figure 19: ACE ablation effect on SIF. The motion-space adversarial constraint is the relevant positive mechanism in the evaluated ACE family.

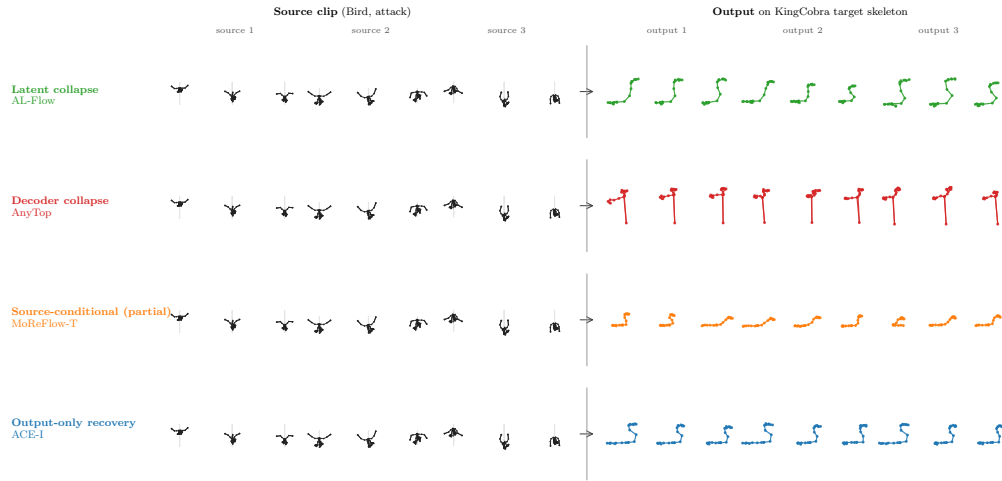


Figure 20: A source-clip substitution view on one Bird $\rightarrow$ KingCobra attack triple. The first three columns, labeled source 1 to source 3, are different Bird attack clips, while the target skeleton and action are fixed; the three columns on the right, labeled output 1 to output 3, show the corresponding KingCobra motions generated by each row. Each row is labeled by its source-instance failure mode: latent collapse for AL-Flow, where source identity is lost before decoding; decoder collapse for AnyTop, where latent variation is not expressed in the output; partial source-conditionality for MoReFlow-T; and output-only recovery for ACE-I, the clearest output-level source dependence among the displayed rows.

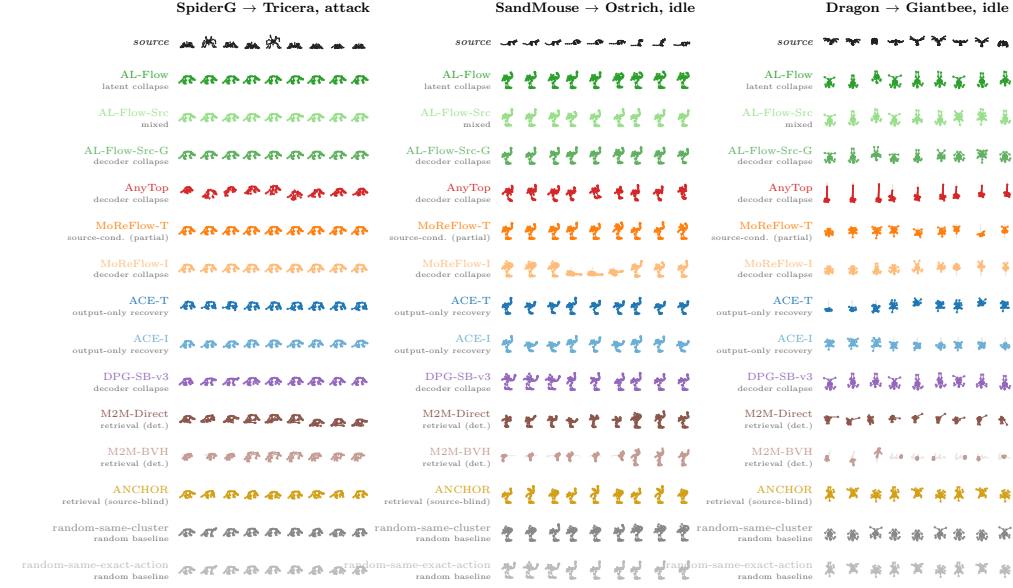


Figure 21: Full qualitative grid A. Three source clips are compared against target outputs from the evaluated methods on a fixed source-target-action triple.

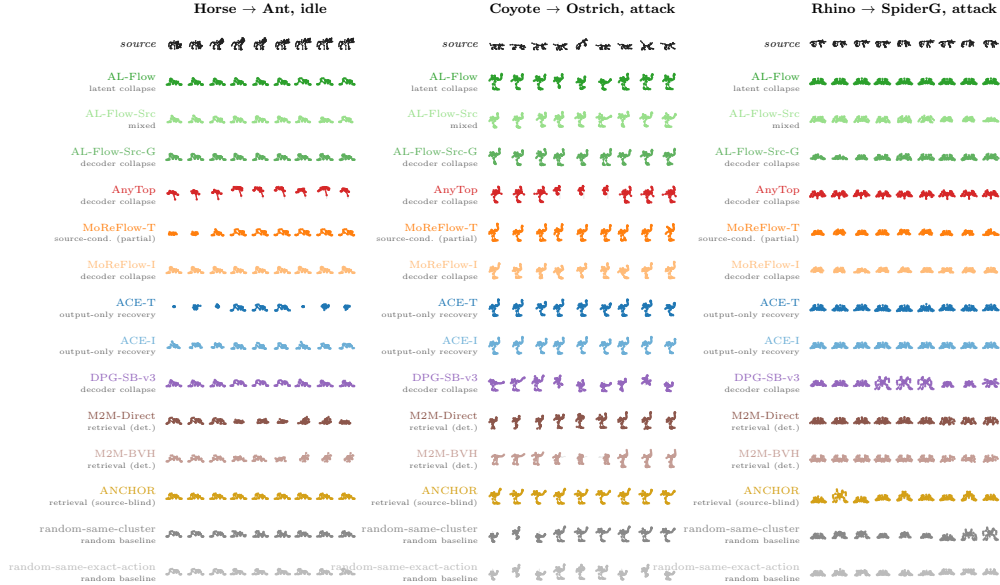


Figure 22: Full qualitative grid B. The same layout is used to reveal whether output variation follows the source clips after target skeleton and action are fixed.

Hippopotamus → Lion, attack



Figure 23: Full qualitative grid C. The grid complements Figure 20 by showing all evaluated methods on an additional held-out triple.

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